

Problem Set 2

POLS W4732

Fall 2026

Due October 2 before the TA section

Instructions. For each formal claim, include two or three algebraic inequalities or best-response comparisons that establish it. Unless directly following notation from slides or the text of the problem, define your notation.

Artificial-intelligence policy. Students may not use generative-AI systems (e.g., ChatGPT, Claude, Gemini, Copilot, etc.) to formulate, derive, solve, explain, or write answers to this problem set. Course materials, ordinary calculators, and collaboration within the registered problem-set group are permitted. **Every group member is responsible for understanding and being able to explain the entire submission.**

Exercise 1. Consider the twice repeated version of the strategic form game below:

	A	B	C
A	9, 9	1, 6	-2, 10
B	6, 1	5, 5	1, 0
C	10, -2	0, 1	2, 2

Let's look at the following strategy:

- Play A in first period
- If (A, A) in first period, play B; otherwise, play B with probability $\frac{1}{6}$ and C otherwise.

Under what conditions both players playing this strategy constitutes a Nash Equilibrium? Compare it to the condition we derived in class.

Exercise 2. Consider the twice repeated version of the Prisoner's Dilemma

1. How many terminal histories does the game have?

Now suppose that we add a period to the game: at $t = 1$, both players simultaneously choose whether entering a revenue-sharing agreement (A) or not (R). If either player chooses R, the payoff from $t = 0$ equals zero and then both players play two regular prisoners' dilemmas in $t = 2$ and $t = 3$. If they both choose A, they enter a binding agreement to split in half the total payoff associated with each of two subsequent periods (so, for example, $\{(D, C), (D, C)\}$ will result in both players obtaining 1.5 in each period). Entering the binding agreement requires that each player pays 0.5 upfront to a third party enforcer (Don Falcone—again).

2. Describe the set of terminal histories of this game and compute the associated average discounted payoffs.
3. Under what conditions is there a Nash Equilibrium in which both players choose A in the first stage?

Exercise 3. Consider the infinitely repeated version of the Prisoner's Dilemma. We characterize conditions under which playing "Tit-for-tat" is a Subgame Perfect Equilibrium

1. No one shot deviations following histories ending in (D, C) requires $\delta \geq 1/2$
2. No one shot deviations following histories ending in (D, D) requires $\delta \leq 1/2$

Argue that both playing Tit-for-Tat is a SPE if and only if $\delta = \frac{1}{2}$

Exercise 4. Consider the infinitely repeated version of the following game G , in which players have discount factor $\delta \in (0, 1)$.

		Pl. 2		
		A	B	C
Pl. 1	A	4, 4	3, 0	1, 0
	B	0, 3	2, 2	1, 0
	C	0, 1	0, 1	0, 0

1. What is the unique NE of the game G ?
2. What is each player's minmax payoff?
3. Let us try to sustain payoff $(2, 2)$. Specifically, consider the following strategy: if the history of play features (B, B) in every period, play B. Otherwise, play C forever. Is this a subgame perfect equilibrium?
4. Consider the following strategy:

- (a) if in the last period the outcome was (B, B) , play B
- (b) otherwise, play C for three periods and, if your opponent did the same, return to B after that
- (c) otherwise, keep playing C

Characterize conditions under which this strategy profile is a subgame perfect equilibrium (or show that it is not) and compare them to the “two-period punishment” strategy that we saw in class

Exercise 5. Consider the infinitely repeated alternating offers game (AOG_∞). Suppose that (i) players are perfectly patient (so $\delta = 1$), but (ii) after rejection, with probability $p > 0$ negotiation terminates, with payoffs (b_1, b_2) . Show that the game has a unique Subgame Perfect Nash Equilibrium

$$v_1 = \frac{1 - b_2 + (1 - p)b_1}{2 - p}$$

$$v_2 = \frac{1 - b_1 + (1 - p)b_2}{2 - p}$$